NET731 – Networking

Week 7 Practical: Monitoring Setup

17–23 April | Total Marks: 10

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Azure Subscription ID:	Subscription ID ↑↓ da8112ec-3211-41b2-835b-f0364e0056fd

Learning Outcomes

By the end of this practical, you should be able to:

- Enable and configure Azure Monitor on virtual machine and network resources
- Navigate the Azure Monitor interface to locate and interpret key metrics
- Identify normal versus abnormal performance baselines for CPU, network, and disk
- Produce a structured monitoring report with supporting evidence

Overview

This week you will configure Azure Monitor across the cloud environment you have been building throughout the semester. Azure Monitor is the central platform for collecting, analysing, and acting on telemetry from your Azure resources. Effective monitoring is not simply about enabling tools — it requires understanding what you are measuring and why it matters.

You are expected to make decisions, explore the interface, and think critically about what you observe. Screenshots must clearly show your own Azure environment — generic or placeholder images will not be accepted.

Before You Begin — Prerequisites

Ensure your VM from previous weeks is running and accessible.
Your Virtual Network (VNet) and Network Security Group (NSG) from Week 5 should be in place.
You will need Contributor access to your Azure resource group.
Allow 10–15 minutes for monitoring data to populate after enabling Azure Monitor.

Task 1 — Enable Azure Monitor (4 Marks)

Azure Monitor needs to be enabled on individual resources before it can collect data. This task covers both your virtual machine and the network resources in your environment.

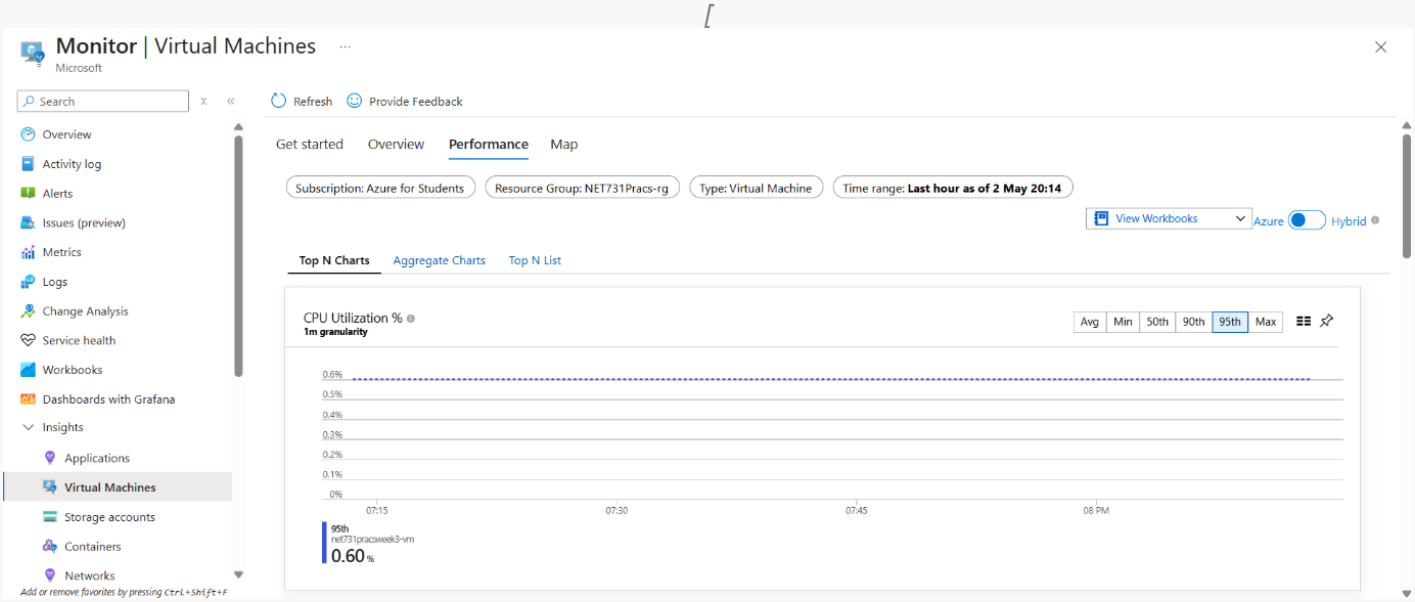
Task 1.1 — Enable Monitoring on Your Virtual Machine (2 Marks)

Navigate to your virtual machine in the Azure Portal and locate the Monitoring section in the left-hand menu. You will find several sub-options here — explore them before enabling anything.

> **Think Before You Click**
> What is the difference between 'Insights' and 'Diagnostics settings' for a VM?
> What data will be collected once you enable monitoring? Where is it stored?
> Does enabling monitoring have any cost implications you should be aware of?

Once you have explored the options, enable VM Insights or Diagnostics for your virtual machine. Configure it to collect at minimum: CPU metrics, memory (if supported), and disk I/O.

[



Screenshot: Azure Monitor enabled on VM — show the Monitoring blade confirming activation]

Insert screenshot here

Describe in 2–3 sentences what you enabled and where the collected data will be stored:

Your response: **What I enabled is Monitor Insight for my virtual machine to track the usage of the specifications such as the CPU, memory and disk space. With this I can track the usage data metrics**

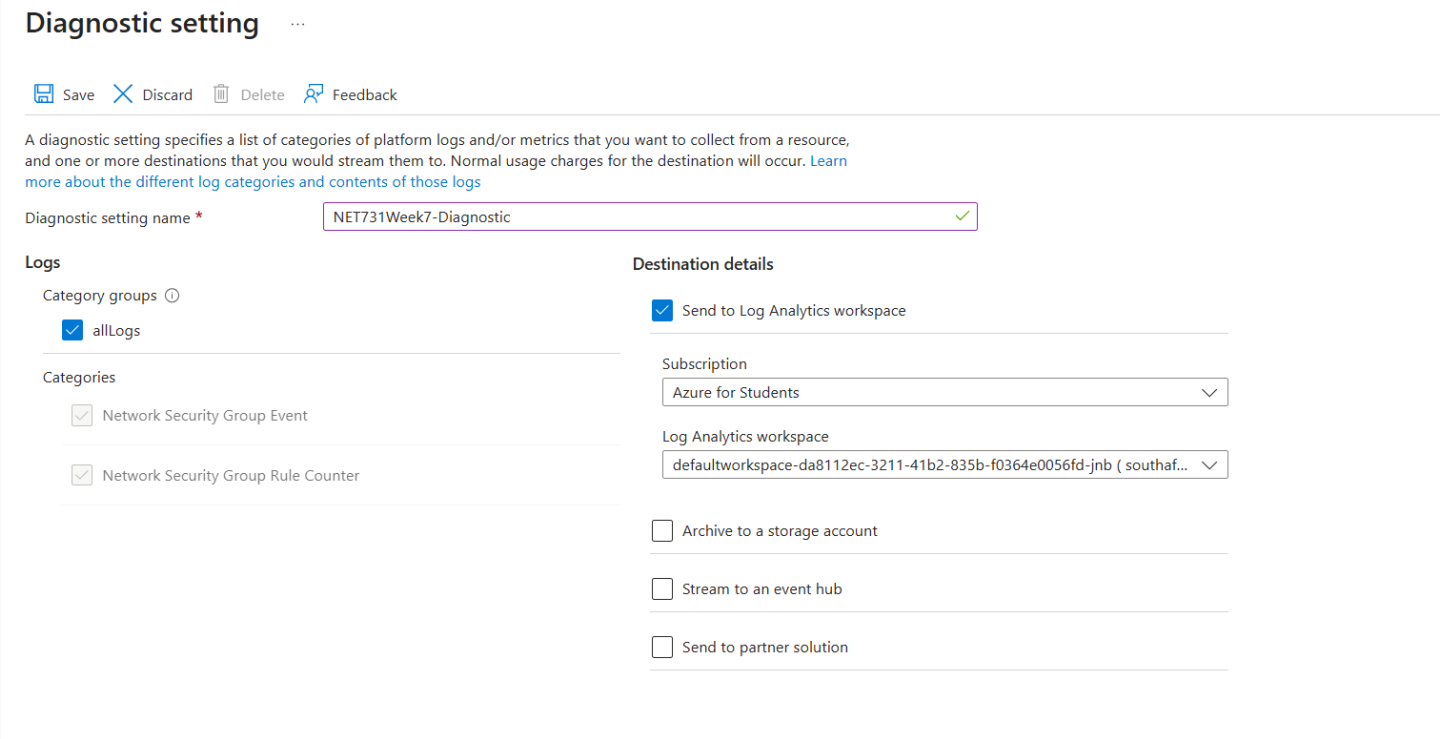
and gauge if I need to vertically upscale my virtual machine. This data will then be stored in Azure Log Monitor Log Analytics workspace which is a central repository for log and performance data.

Task 1.2 — Enable Monitoring on Network Resources (2 Marks)

Repeat the process for your network resources. Both your NSG and your VNet support diagnostic settings.

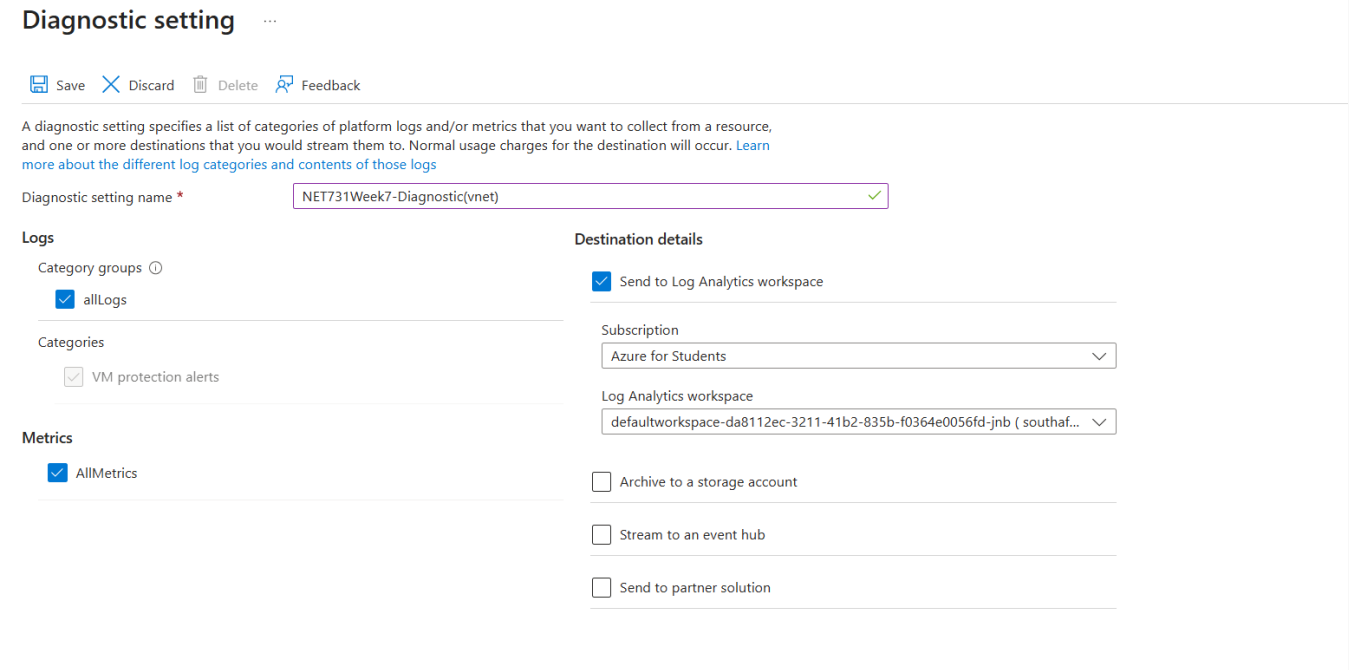
Consider the following before proceeding:

- What specific log categories are available for an NSG? Which ones would be most useful in a security context?
- What is a Log Analytics Workspace, and why might you need one to proceed?
- If you do not already have a Log Analytics Workspace, you will need to create one — consider which region is appropriate.



[Screenshot: Diagnostic settings enabled on NSG — show categories selected and destination]

Insert screenshot here



[Screenshot: Diagnostic settings on VNet or additional network resource]

Insert screenshot here

Task 2 — Review Key Metrics (3 Marks)

Now that monitoring is enabled, you will use Azure Monitor's Metrics Explorer to observe and interpret the data being collected. Generating some load on your VM before taking screenshots will produce more meaningful graphs.

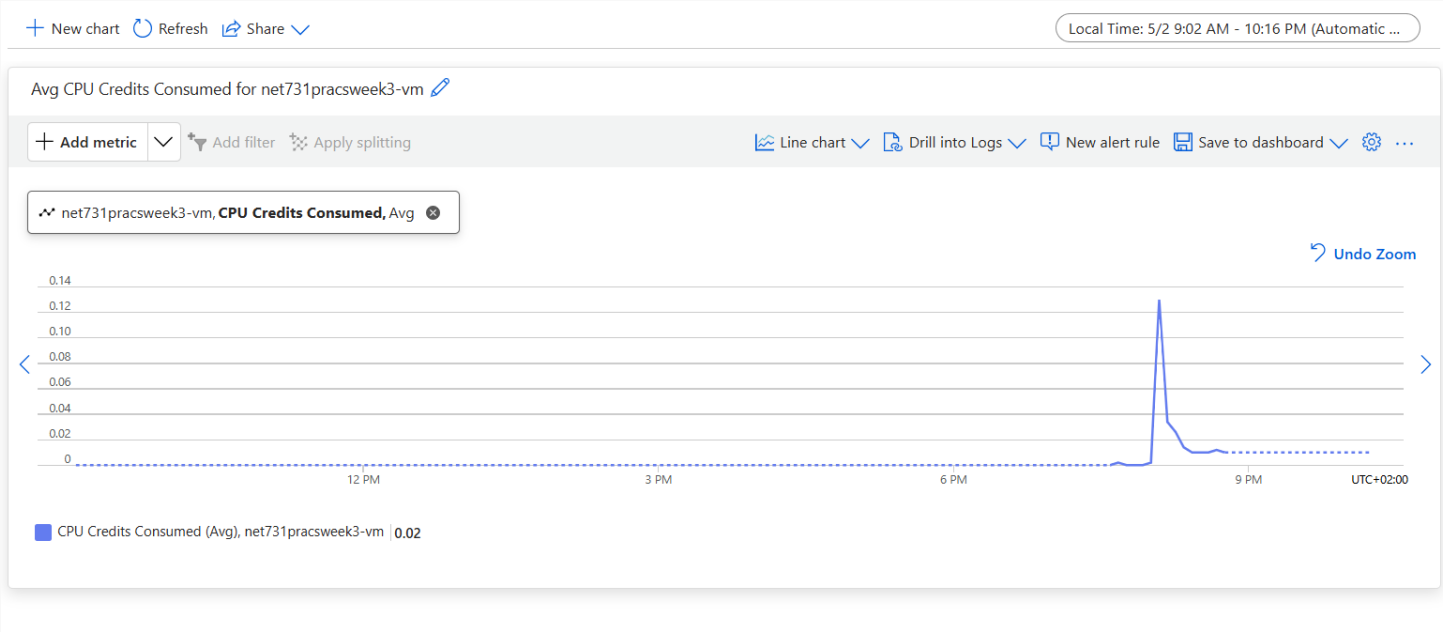
Generating Test Load (Optional but Recommended)

On Linux VMs: the 'stress' tool can simulate CPU and I/O load.
On Windows VMs: Task Manager's 'New Task' can run a CPU-intensive process.
Even browsing the web or installing packages from within the VM will generate measurable network activity.
Allow a few minutes after generating load before capturing screenshots.

Task 2.1 — CPU Metrics (1 Mark)

Open Azure Monitor > Metrics and select your VM as the scope. Explore the available metrics before choosing one to display.

- Select an appropriate time range that shows meaningful data (consider: what time range makes sense for a lab exercise?)
- Identify which metric reflects actual CPU usage — consider why there might be more than one CPU-related metric
- Add annotations or use the pin feature to highlight any interesting patterns you observe



[Screenshot: CPU metrics chart — VM scope, showing usage over a meaningful time period]

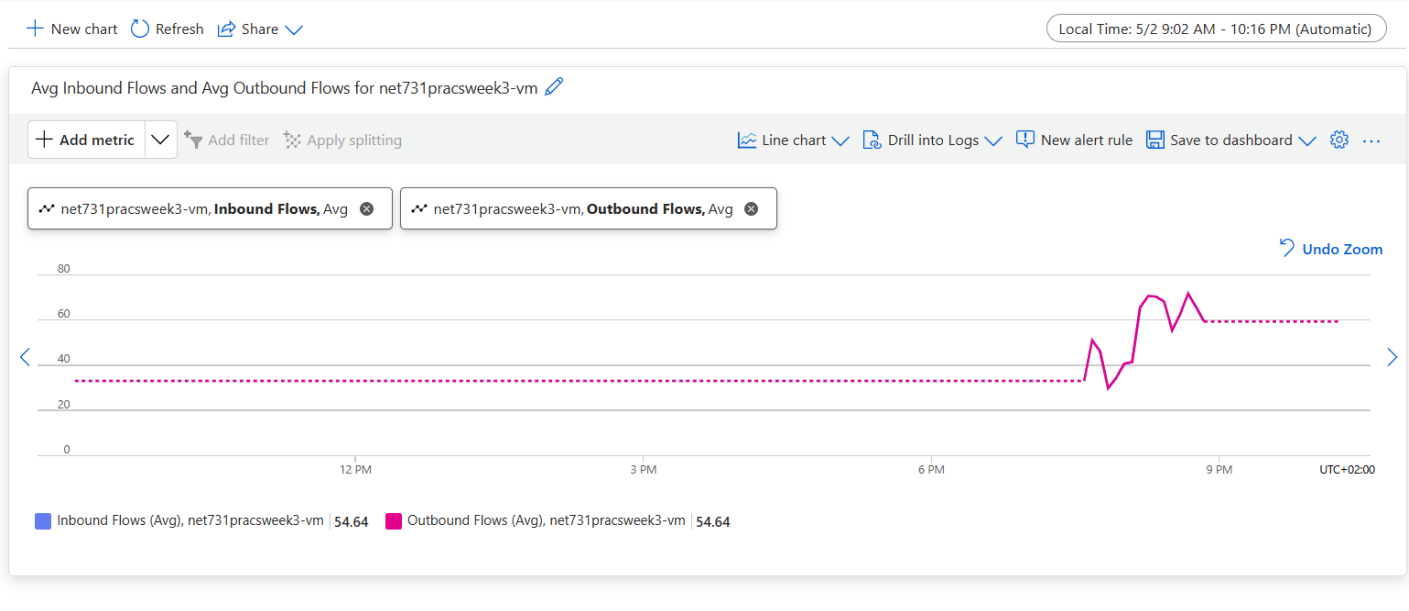
Insert screenshot here

In 1–2 sentences, describe what the CPU metric is showing and whether the values appear expected for your workload: **The CPU metrics shown are for CPU credits consumed. Yes, they are accurate to the workload because I just turned my VM on a couple of minutes ago.**

Task 2.2 — Network Metrics (1 Mark)

Still in Metrics Explorer, change the metric to observe network throughput. You should investigate both inbound and outbound traffic.

- What is the unit of measurement for the network metric you selected? Is it what you expected?
- Can you display both Network In and Network Out on the same chart? How would you do this?



[Screenshot: Network metrics chart — showing inbound and/or outbound traffic]

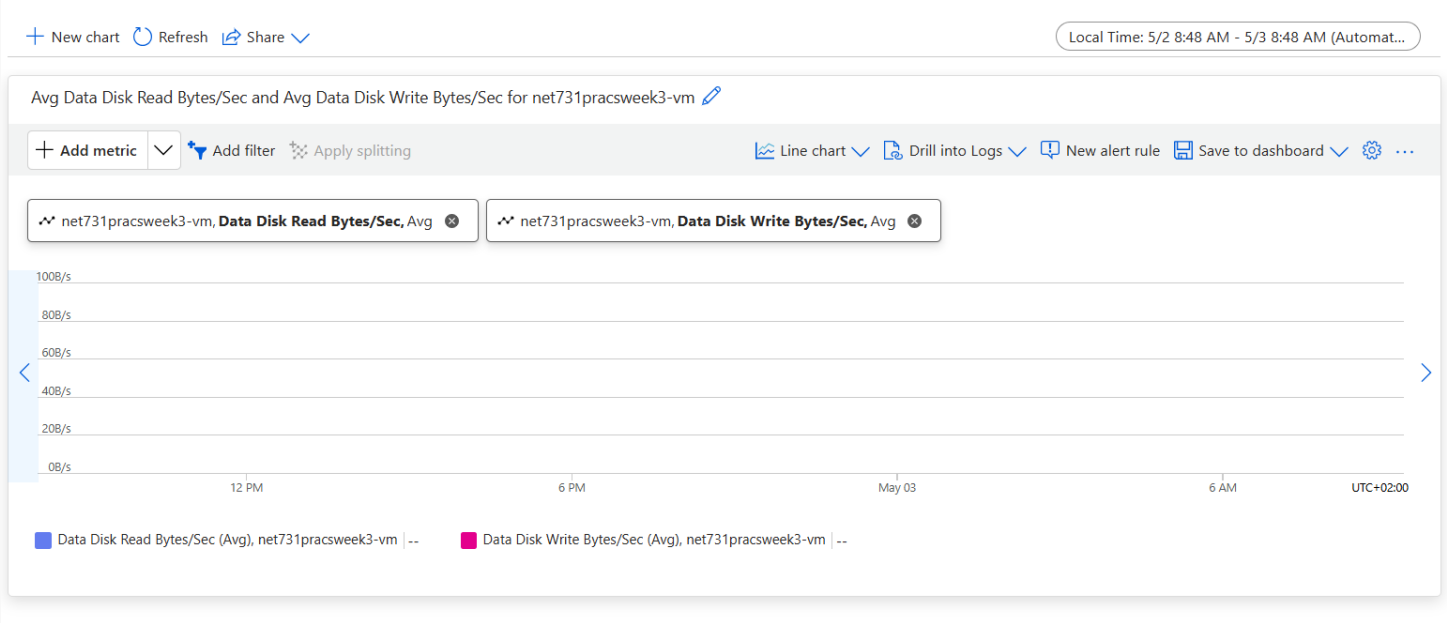
Insert screenshot here

Note any observations about the network traffic pattern you can see: **There is not particularly any inbound flows due to the virtual machine not being used to receive data.**

Task 2.3 — Disk Metrics (1 Mark)

Finally, explore the disk-related metrics for your VM. Azure exposes separate metrics for disk reads and disk writes.

- Locate both Disk Read Bytes and Disk Write Bytes (or equivalent metrics in your VM configuration)
- Consider: during what activities would you expect disk write activity to be high?



[Screenshot: Disk metrics chart — showing read and/or write activity]

Insert screenshot here

Describe in 1–2 sentences what the disk metrics reveal about your VM's activity: **The write metric on the VM activity measures the speed, volume and latency of the data being saved from the virtual machine to its storage disk. The disk read metric measures how much data is collected from the storage to identify where there are bottlenecks.**

Task 3 — Document Your Monitoring Setup (3 Marks)

Monitoring documentation is a critical deliverable in professional environments. A future administrator — or your future self — should be able to understand your setup from the document alone.

Task 3.1 — Monitoring Summary (2 Marks)

Write a structured summary of your monitoring configuration. Your summary should cover all three areas below. Aim for clarity and precision — this is professional technical documentation.

Resources Monitored

List each Azure resource you have enabled monitoring on, specifying the monitoring method used (e.g., VM Insights, Diagnostic Settings) and what data each resource is collecting:

Answer: The resources that I enabled monitoring on are my virtual machine, NSG and VNet. For the Virtual machine, what I enabled is the virtual machine insights to be able to track the usage metrics and performance whenever someone uses the virtual machine. For the NSG and the virtual network, I enabled the diagnostic

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Data Retention and Storage

Where is your monitoring data stored? What retention period applies? How would you access historical data older than the default retention window?

The monitoring data is stored in the Azure Log Monitor Log Analytics workspace which is a central repository for log and performance data. The retention period in my Log analytics workspace is 30 days. To access historical data that is older than the retention period, I would collect it from the archived data that remains in the azure environment, but the data would need to be restored first.

Metric Observations

Summarise what you observed across the three metrics reviewed in Task 2. Were any values unexpected? What might cause anomalies in these metrics in a production environment?

The Answer: About the metrics that I was observing in task 2, there were no anomalies or data that I thought was inaccurate. The only data metric that was interesting was the metrics for the inbound and outbound traffic for the VM, being that the inbound traffic metric didn't contain any data because as I said before, the virtual machine has not been used therefore there will not be any data that shows that it received traffic from other sources.

Task 3.2 — Extension: Configure an Alert Rule (1 Bonus Mark)

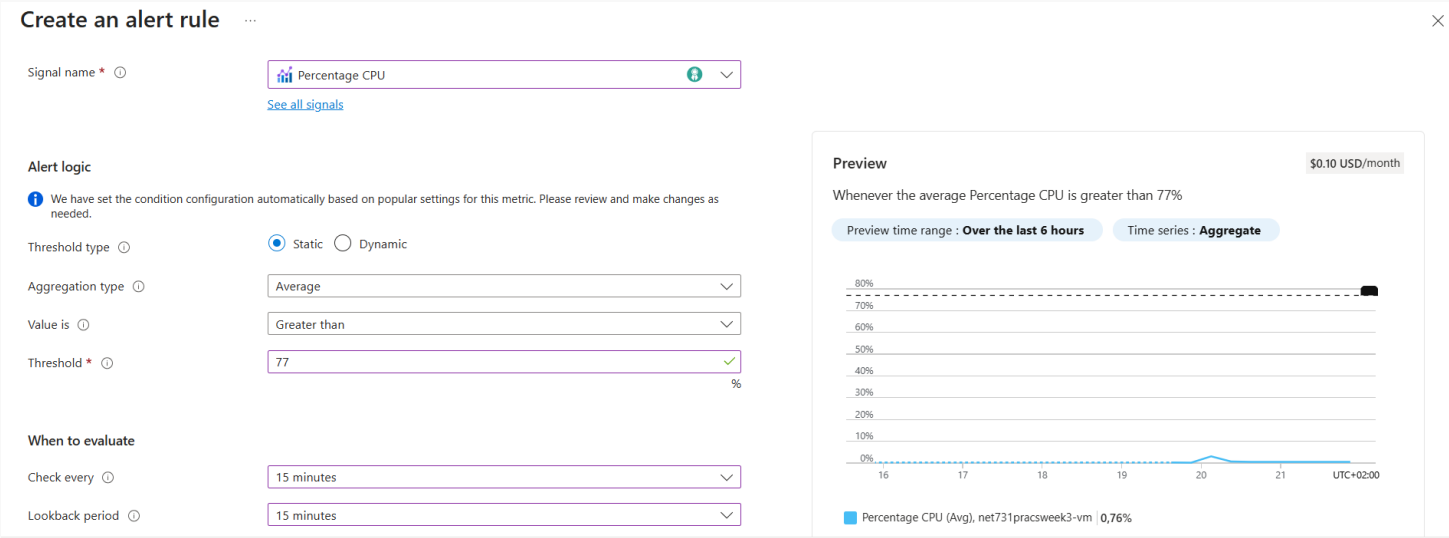
Extension Task

This task is optional but provides an opportunity to demonstrate deeper understanding.

Configure at least one Azure Monitor Alert Rule on your VM or network resource.

The alert should have a meaningful threshold — for example, triggering when CPU exceeds 80% for 5 minutes.

Document your alert logic: what condition triggers it, what action does it take, and why you chose that threshold.



[Screenshot: Configured alert rule — show condition, threshold, and action group]

Insert screenshot here

Explain the alert rule you configured and the reasoning behind your chosen threshold**: The Alert rule is just made is for the CPU usage. What is does is that it will notify the Owner of the virtual machine when the CPU usage exceeds 77% which is the percentage I set. I chose this specific alert for the case of testing the virtual machine's testing to see if the workload that will be sufficient for the virtual machine so if it exceeds 77%, I know that we need to get a much larger size to deploy so it can run smoother.**

Reflection Questions

Answer these questions thoughtfully. There are no single correct answers — markers are looking for evidence of understanding and critical thinking.

Reflection Question	Your Response
Why is monitoring important in a cloud environment? How does it differ from monitoring on-premises infrastructure?	Monitoring is important in a cloud environment because it helps azure owners secure and evaluate resources and their usages which can save costs. It is different to on-premises protection due to the fact that it online and not on physical servers.

What is the difference between a metric and a log in Azure Monitor? When would you use each?	A metric are statistics of a resource and logs are actions that we performed in resources.
If your VM showed consistently high CPU usage (above 90%) every day at 2am, what steps would you take to investigate?	I would implement alerts and if it continues, I would deploy a VM with a much larger CPU.
How could monitoring data be used to make cost-optimisation decisions in Azure?	Monitoring resources can track specific aspect metrics that are being used and how frequently they are being used, and it cloud help make smarter decisions on what types of resources to deploy and be able to make precise purchases and save money

Marking Criteria

Use this table to check your submission before submitting.

Task	Marks	Evidence Required
1.1 Enable Azure Monitor on VM	**2**	*Screenshot of Monitor enabled on VM resource*
1.2 Enable Monitoring on Network Resources	**2**	*Screenshot showing NSG/VNet diagnostics*
2.1 CPU Metrics Review	**1**	*Screenshot of CPU usage chart with annotations*
2.2 Network Metrics Review	**1**	*Screenshot of network in/out metrics*
2.3 Disk Metrics Review	**1**	*Screenshot of disk read/write metrics*
3.1 Monitoring Documentation	**2**	*Written summary with all screenshots included*
3.2 Alert Configuration (Extension)	**1**	*Screenshot of configured alert rule*
TOTAL	**10**	

Submission Checklist

All screenshot placeholders have been replaced with real screenshots from your Azure environment.
Each screenshot clearly shows your resource names and Azure subscription context.
Written responses are your own words — not copied from documentation.
Reflection questions have been answered with evidence of understanding.
The document has been saved as a .docx or .pdf and submitted via the course portal.

Useful References

The following official documentation may help — but always explore the Azure Portal first before consulting documentation:

- Azure Monitor overview — docs.microsoft.com/azure/azure-monitor/overview
- Enable VM Insights — docs.microsoft.com/azure/azure-monitor/vm/vminsights-enable-overview
- Metrics Explorer — docs.microsoft.com/azure/azure-monitor/essentials/metrics-getting-started
- Diagnostic settings — docs.microsoft.com/azure/azure-monitor/essentials/diagnostic-settings
- Azure Monitor Alerts — docs.microsoft.com/azure/azure-monitor/alerts/alerts-overview

Tip: The Azure Portal's built-in 'Learn more' links on each blade are often the fastest route to relevant documentation.